

DevAtlas System Architecture and Operations Guide

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1. Main system flow

DevAtlas helps engineering teams manage service documents and incident knowledge. After login, a user creates a knowledge base and uploads TXT, Markdown, or PDF files. The backend validates the file, stores the original, extracts text, splits it into chunks, creates embeddings, writes vectors to Chroma, and finally marks the version as indexed.

When a user asks a question, the server validates the JWT and knowledge-base ownership. It embeds the question, searches the current knowledge base with metadata filters, and retrieves the Top-K chunks. The chunks and question are sent to DeepSeek. The answer is delivered to the browser through SSE events named token, citation, done, and error.

2. Important document states

State	Meaning	User action
pending	The file is being parsed, split, or indexed.	Wait for processing.
indexed	Vectors are in Chroma and can be retrieved.	Start asking questions.
failed	A pipeline step failed and an error was saved.	Read error_message and reindex.

3. Authorization model

Every knowledge base has an owner_id. Read, upload, delete, question, and incident endpoints must check both the authenticated user and the knowledge-base ID. Guessing a numeric URL ID must never expose another user's documents or incident records.

4. Document versions and fingerprints

A logical document represents a long-lived identity, while a version is one content snapshot. Upload processing calculates a SHA-256 fingerprint. The same fingerprint returns duplicate=true and avoids an unnecessary version. A changed fingerprint creates a new version_number. SHA-256 is a content fingerprint, not encryption.

5. RAG quality checks

Top-K=3 means at most three similar chunks are retrieved; it does not limit the answer to three sentences. A distance value represents vector distance and is normally smaller for more similar text. Check the cited filename, chunk index, and distance. If the knowledge base has no evidence, the assistant should say that the evidence is insufficient instead of inventing configuration.

6. SSE events

SSE uses a text/event-stream connection for one-way server-to-browser delivery. A token event carries a partial answer, citation carries source metadata, done means normal completion, and error means the service failed. Chunked arrival is normal; the frontend parses event boundaries and appends token text in order.

7. Incident analysis

An incident contains a title and logs or symptoms. The service uses the same retrieval chain to generate causes, investigation steps, and risk notes, then saves input, result, status, and citations. completed means success, failed means an exception occurred, and cancelled means the user stopped the stream.

8. Security and operations

Uploads must enforce an extension allowlist and size limit, normalize filenames, and block path traversal. Database passwords, JWT secrets, and DeepSeek API keys belong in environment variables. Logs must not print complete tokens. Deleting a document must remove the original file, database versions, and Chroma vectors so old content cannot be retrieved.